

Laurent Colbois

AI/ML EXPERT · PHD IN MACHINE LEARNING & BIOMETRICS

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“To create a positive impact through technology while following sustainable practices.”

Summary

Machine Learning expert with PhD from UNIL/Idiap, passionate about enabling AI adoption through support, training, and knowledge transfer. Specialized in bridging technical complexity and diverse audiences—from teaching staff to researchers to administrative teams. Proven ability to mentor students, coordinate cross-functional projects, and translate AI concepts for non-technical stakeholders. Seeking to leverage expertise in a central coordination and support role that empowers others to effectively use AI tools and methods.

Work Experience

Idiap Research Institute

Martigny, Switzerland

POSTDOCTORAL RESEARCHER - VISION-LANGUAGE MODELS FOR BIOMETRICS

Apr. 2025 - Mar. 2026

- Benchmarked open-weight Vision-Language Models (Gemma3, Qwen2.5-VL, InternVL3) for explainable face recognition, developing methodologies to evaluate AI explanation quality and trustworthiness
- Built reproducible ML pipelines using modern MLOps tools (Hydra, Weights & Biases, vLLM) and worked with HPC infrastructure (Slurm cluster, GPU computing) for efficient AI model deployment
- Led cross-functional project coordination for a forensic data collection platform, managing communication between researchers, developers, and legal staff while translating technical requirements into clear specifications
- Won first place (15K CHF) at 9-day startup hackathon by developing AI-based music generation demo and successfully presenting technical concepts to diverse audiences (technical experts, business stakeholders, investors)

Idiap Research Institute

Martigny, Switzerland

PHD STUDENT - DEEPFAKES, BIOMETRICS, FORENSIC SCIENCE

Jan. 2021 - Mar. 2025

- Supervised and mentored four master's students on score calibration projects for forensic face recognition, providing technical guidance, problem-solving support, and research methodology training
- Taught annual course on forensic evaluation of face comparison tasks, presenting complex AI concepts to forensic science students and practitioners
- Delivered outreach talks on AI and deepfakes to forensic science community, police forces, and general public—adapting technical content to audience knowledge level and building trust through clear, accessible communication
- Developed and maintained reproducible research code packages emphasizing workflow automation, experiment tracking, and code quality—serving as internal resource for lab members on best practices

Idiap Research Institute

Martigny, Switzerland

RESEARCH INTERN - SYNTHETIC DATA FOR FACE RECOGNITION

Jul. - Dec. 2020

- Created and published synthetic face recognition dataset using generative AI (GANs) for benchmarking purposes

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Lausanne, Switzerland

RESEARCH INTERN - AUDIO MACHINE LEARNING

Feb. - Aug. 2019

- Developed automated audio scene classification model for auto-equalization and implemented real-time demo on embedded microcontroller

Education

University of Lausanne / Idiap Research Institute

Lausanne, Switzerland

PHD IN FORENSIC SCIENCE (MACHINE LEARNING & BIOMETRICS)

2021 - 2025

- Thesis: Face morphing attacks in the era of deepfakes: risks, detection & source attribution

- Thesis: Trainable filterbanks for end-to-end speech enhancement via time-frequency masking
- Skills: Numerical simulations, Machine learning, Deep learning, Signal processing, High performance computing

- Skills: Applied mathematics, Numerical physics, Programming, Scientific computing

1 Publications

2026	Conference Paper (Under Review) , Benchmarking the Explanatory Quality of Open-Weight Vision-Language Models in Face Recognition	<i>Face and Gesture</i>
2024	Conference Paper , Evaluating the effectiveness of attack-agnostic features for morphing attack detection	<i>IJCB</i>
2023	Conference Paper , Approximating Optimal Morphing Attacks using Template Inversion	<i>IJCB</i>
2022	Conference Paper , On the detection of morphing attacks generated by GANs	<i>BIOSIG</i>
2022	Conference Paper , Are GAN-based morphs threatening face recognition?	<i>ICASSP</i>
2021	Conference Paper , On the use of automatically generated synthetic image datasets for benchmarking face recognition	<i>IJCB</i>

Skills

Domains	AI/ML consulting & support, Machine Learning, Deep learning, Computer vision, Vision-Language Models, Teaching & mentorship, Cross-functional communication, Project coordination, Reproducible research
ML & Programming	Python, PyTorch, Huggingface Transformers, vLLM, Scikit-Learn, Pandas, NumPy, TensorFlow (basic), C++ (basic)
MLOps & Infrastructure	Git, Docker, Apptainer/Singularity, Conda, Pip, Pixi, Snakemake, Prefect, Weights & Biases, MLFlow, Hydra, Slurm (HPC clusters), Unix/Linux
Languages	French (Mother tongue), English (Professional proficiency), German (B2, reactivation in progress)